MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

GATE, INVERTER, DIGITAL, KEY CODE DBE

INACTIVE FOR NEW DESIGN AFTER DATE OF THIS SPECIFICATION

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic inverter gate module.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.4 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be DBE.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/156-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I(see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 Vdc, except that the limits for output short circuit current and output impedance apply only for a power supply voltage of 5.5 Vdc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

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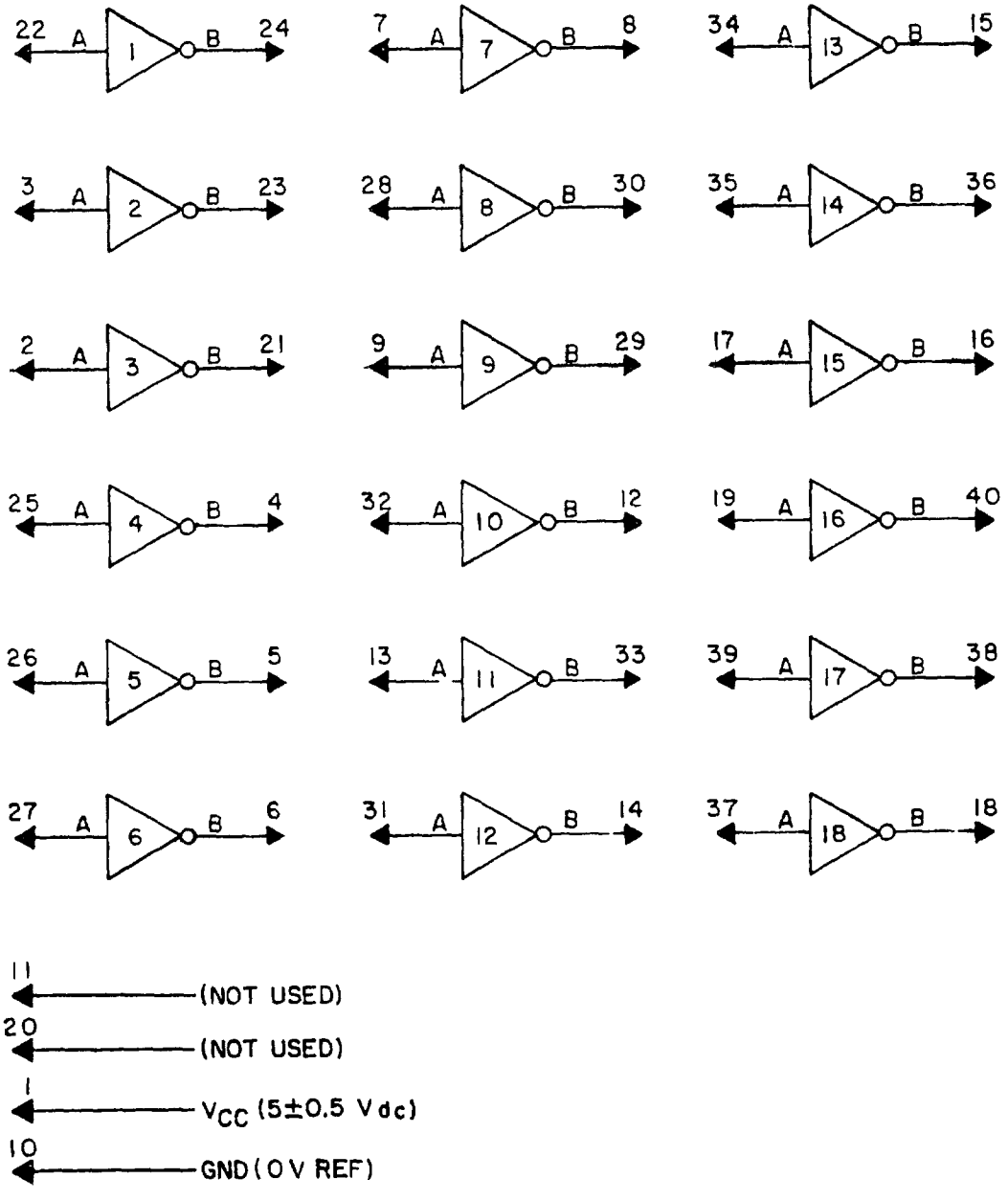


FIGURE 1. DBE module function diagram, inverter gate.

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25° ±5°C Ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A18	V_{OUT} (low-loaded)	III	S	0.0	0.6	0.0	0.6	0.0	0.6	0.0	0.6	V
B1-B18	Input breakdown current	IV	S	0	100	0	100	0	100	0	100	µA
C1	Isolation, frame	V	S	10	∞	1.0	∞	10	∞	1.0	∞	MΩ
C2-C3	Isolation, unused pin	V	X	10	∞	1.0	∞	10	∞	1.0	∞	MΩ
D1-D18	Output voltage, low level	VI	X	0.0	0.3	0.0	0.3	0.0	0.3	0.0	0.3	V
E1-E18	Output voltage, high level	VII	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
F1-F18	Input current, low level	VIII	X	50	200	50	200	50	200	50	200	µA
G1-G18	Input current, high level	IX	X	0	10	0	10	0	10	0	10	µA
H1-H18	Output short circuit current	X	X	3	15	3	15	3	15	3	15	mA
J1	Power supply current, low level	XI	S	1.0	9.2	1.0	9.2	1.0	9.2	1.0	9.2	mA
J2	Power supply current, high level	XI	S	0.4	3.6	0.4	3.6	0.4	3.6	0.4	3.6	mA
K1-K18	Transition time (t_{TLH}), CL = 40 pF	XII	S	5	60	5	60	5	60	5	60	ns
K19-K36	Transition time (t_{TLH}), CL = 140 pF	XII	Q	10	100	10	100	10	100	10	100	ns
L1-L18	Transition time (t_{THL}), CL = 40 pF	XIII	S	2	30	2	30	2	30	2	30	ns
L19-L36	Transition time (t_{THL}), CL = 140 pF	XIII	Q	5	60	5	60	5	60	5	60	ns
M1-M18	Propagation delay time (t_{PLH}), CL = 40 pF	XIV	S	10	60	10	60	10	70	10	70	ns
M19-M36	Propagation delay time (t_{PLH}), CL = 140 pF	XIV	Q	10	90	10	90	10	100	10	100	ns
N1-N18	Propagation delay time (t_{PHL}), CL = 40 pF	XV	S	10	66	10	66	10	70	10	70	ns
N19-N36	Propagation delay time (t_{PHL}), CL = 140 pF	XV	Q	10	96	10	96	10	100	10	100	ns
P1-P18	Output crosstalk, low level	XVI	Q	0	400	0	400	0	400	0	400	mV
R1-R18	Output crosstalk, high level	XVII	Q	0	400	0	400	0	400	0	400	mV
S	Power supply transient amplitude	XVIII	Q	0	300	0	300	0	300	0	300	mV
T	Power supply filter capacitance	XIX	S	1.65	2.75	1.65	2.75	1.65	2.75	1.65	2.75	µF
U1-U18	Input capacitance	XX	Q	2	15	2	15	2	15	2	15	pF
V1-V36	Output impedance, power off, check	XXI	Q	3.5	5.55	3.5	5.55	3.5	5.55	3.5	5.55	V

1/ "X" parameters shall be 100-percent tested during module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (1 percent, or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

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4.4.2 Order of testing. Test conditions A through V shall be performed in any order with the following exceptions. If test conditions A, B, or C are part of a sequence of tests, the order of testing shall be as follows. A, B, C where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through V in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through T shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests D through T shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests D through T shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Cross-reference of classification. Part number M28787/156-1 supersedes part number M28787/156. The two parts are interchangeable.

6.4 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table I.

Maximum power dissipation	
All outputs at low level	50 mW (outputs not loaded)
All outputs at high level	20 mW (outputs not loaded)
Supply voltage	5 Vdc ±10 percent
Input voltage	
Low level	-0.5 V minimum, 0.7 V maximum
High level	2.0 minimum, V_{CC} maximum

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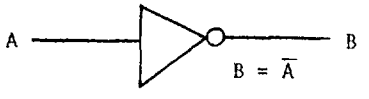
Input current	
Low level (V_{IL} = 0.0 V)	-200 µA maximum
(V_{IL} = 0.3 V)	-180 µA maximum
High level (V_{IH} = 2.4 V)	10 µA maximum
(V_{IH} = 5.5 V)	100 µA maximum
Output voltage	
Low level (I_{OL} = 2.0 mA)	0.3 V maximum
High level (I_{OH} = 100 µA)	2.4 V minimum
Output drive current	
Maximum capability	
Low level (V_{OL} = 0.3 V)	2.0 mA
High level (V_{OH} - 2.4 V)	-100 µA

6.5 Module function. The module contains 18 independent low power inverter gates.

6.5.1 Input to output relationship. Each inverter has the input to output relationship of table II for positive logic.

TABLE II. Input to output relationship.

Input	Output
A	B
L	H
H	L



H = High level.
L = Low level.

6.5.2 Parallel outputs. Paralleling of outputs to obtain extended drive is permissible only when parallel inputs and outputs are confined within each group specified below. The output drive capability of this combination is equal to the sum of the output drive capabilities of the individual circuits. The input loading of this combination is equal to the sum of the input loading of the individual circuits.

The circuit groups which can be paralleled are.

a. Circuits 1, 2, 3, 4, 5, and 6.

b. Circuits 7, 8, 9, 10, 11, and 12.

c. Circuits 13, 14, 15, 16, 17, and 18.

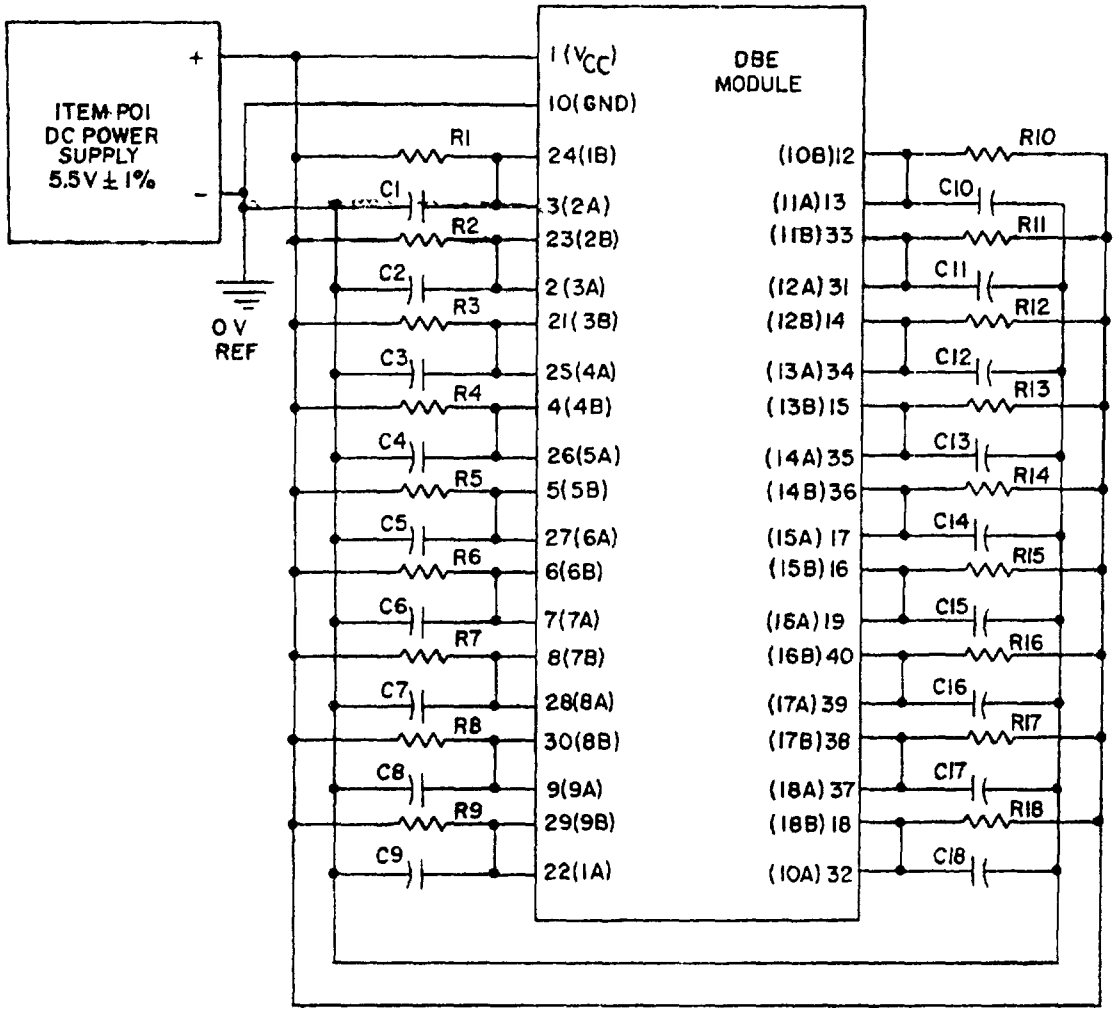
6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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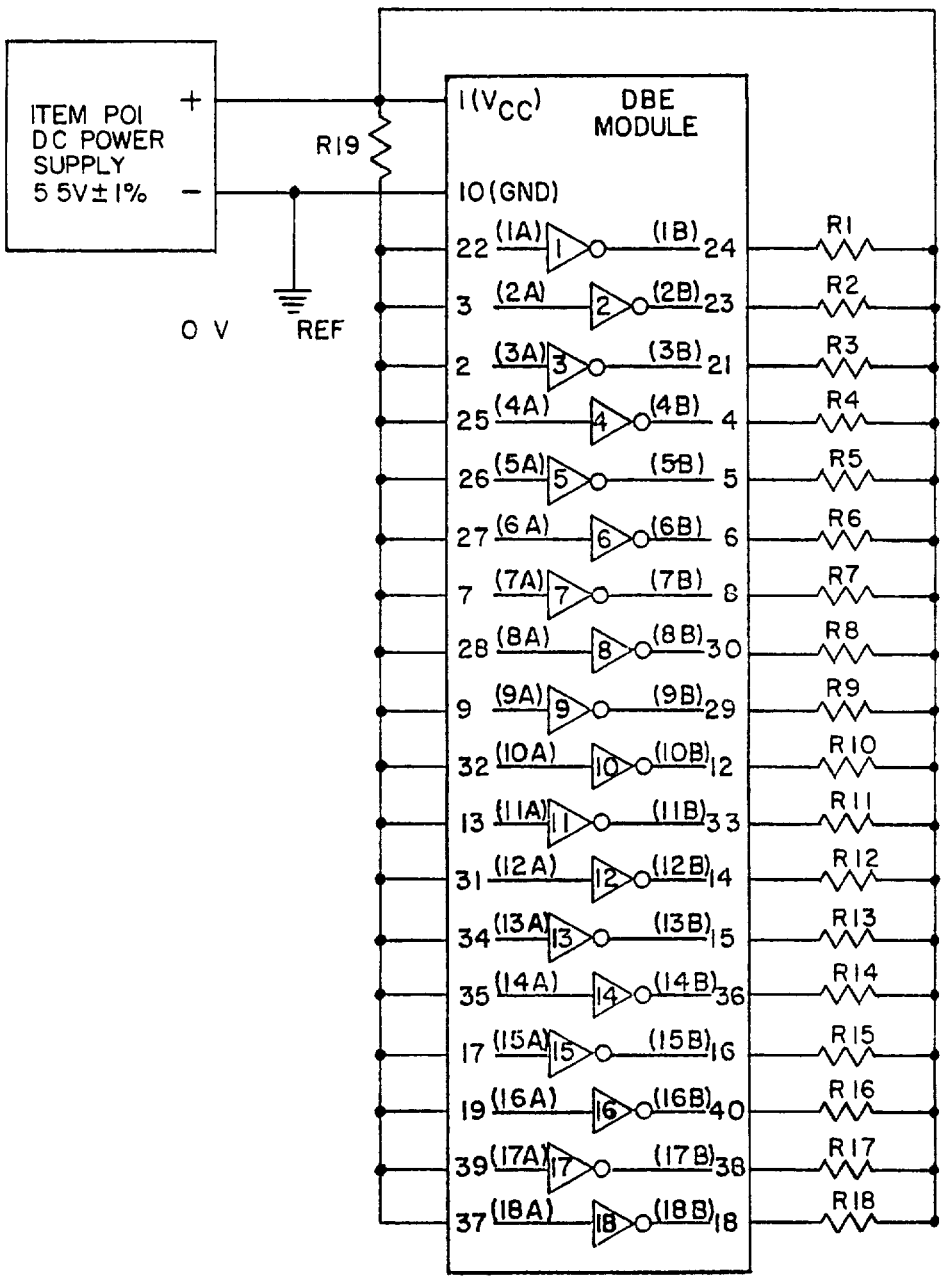
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NOTES:
1. R1 - R18 = 2.75 kΩ ±1%, 1/8 W.
2. C1 - C18 = 100 pF ±20%.

FIGURE 2. Operating temperature and life test circuit.

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NOTE: R1-R19 = 2.75 kΩ ±1%, 1/8 W.

FIGURE 3. Component temperature test circuit.

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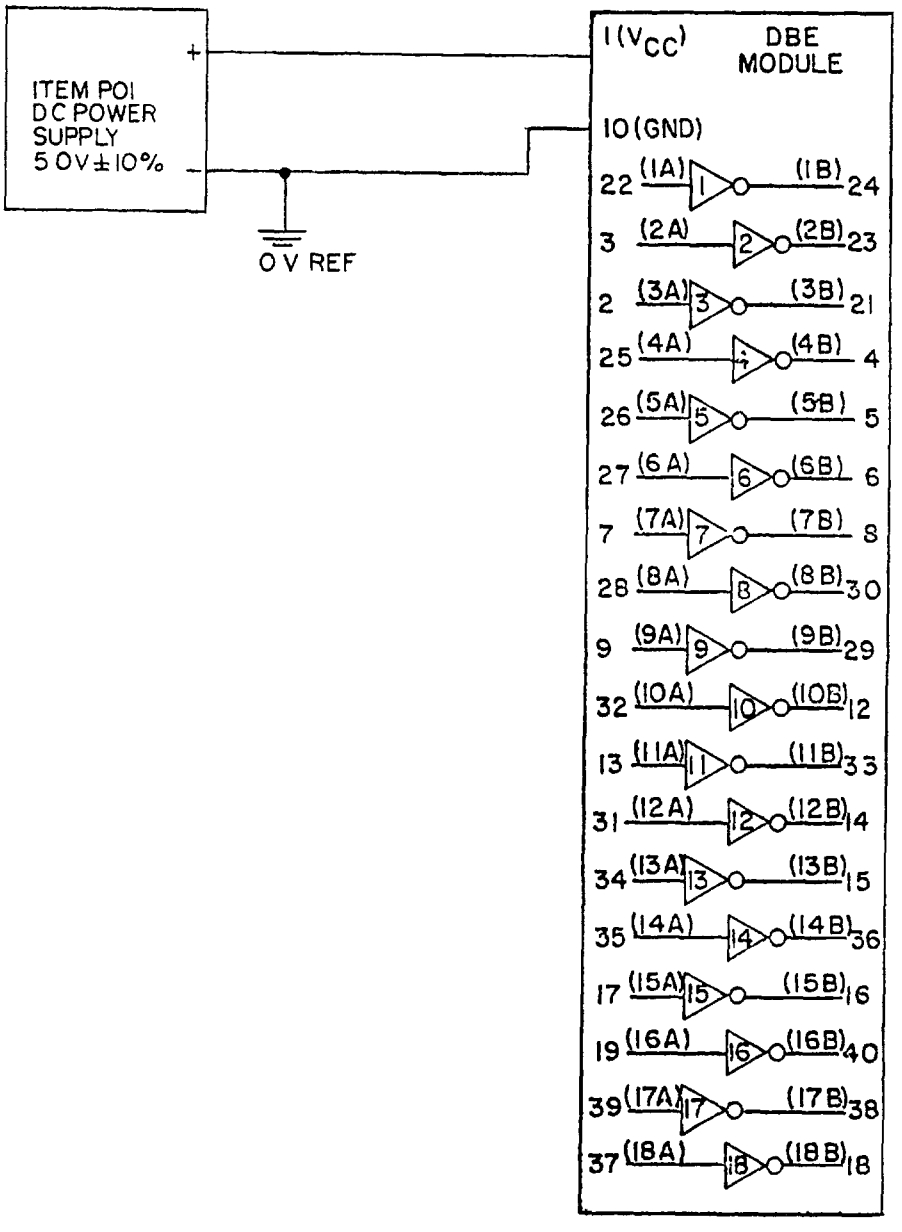


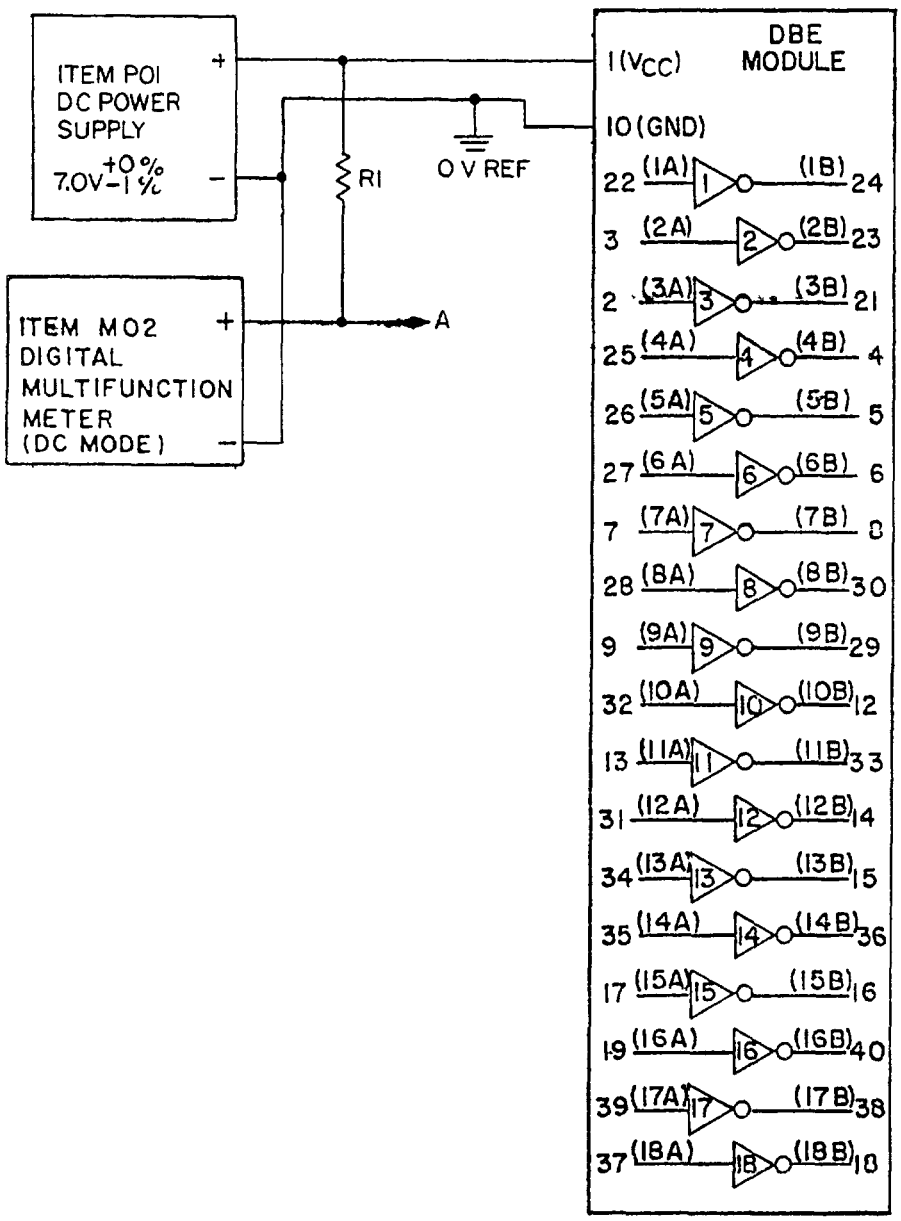
FIGURE 4. Vibration test circuit.

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TABLE III. V_{OUT} (low-loaded) test.

Test condition	Parameter	Connect	Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
A1	1B V_{OUT} (low loaded)	24	0.0	0.6	V	5
A2	2B	23	↓	↓	↓	↓
A3	3B	21				
A4	4B	4				
A5	5B	5				
A6	6B	6				
A7	7B	8				
A8	8B	30				
A9	9B	29				
A10	10B	12				
A11	11B	33				
A12	12B	14				
A13	13B	15				
A14	14B	36				
A15	15B	16				
A16	16B	40				
A17	17B	38				
A18	18B	18				

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NOTES:
1. R1 = 2.75 kΩ ±1%, 1/8 W.
2. Measurements shall be recorded after point "A" has been connected to the appropriate module pin for at least 5 seconds.

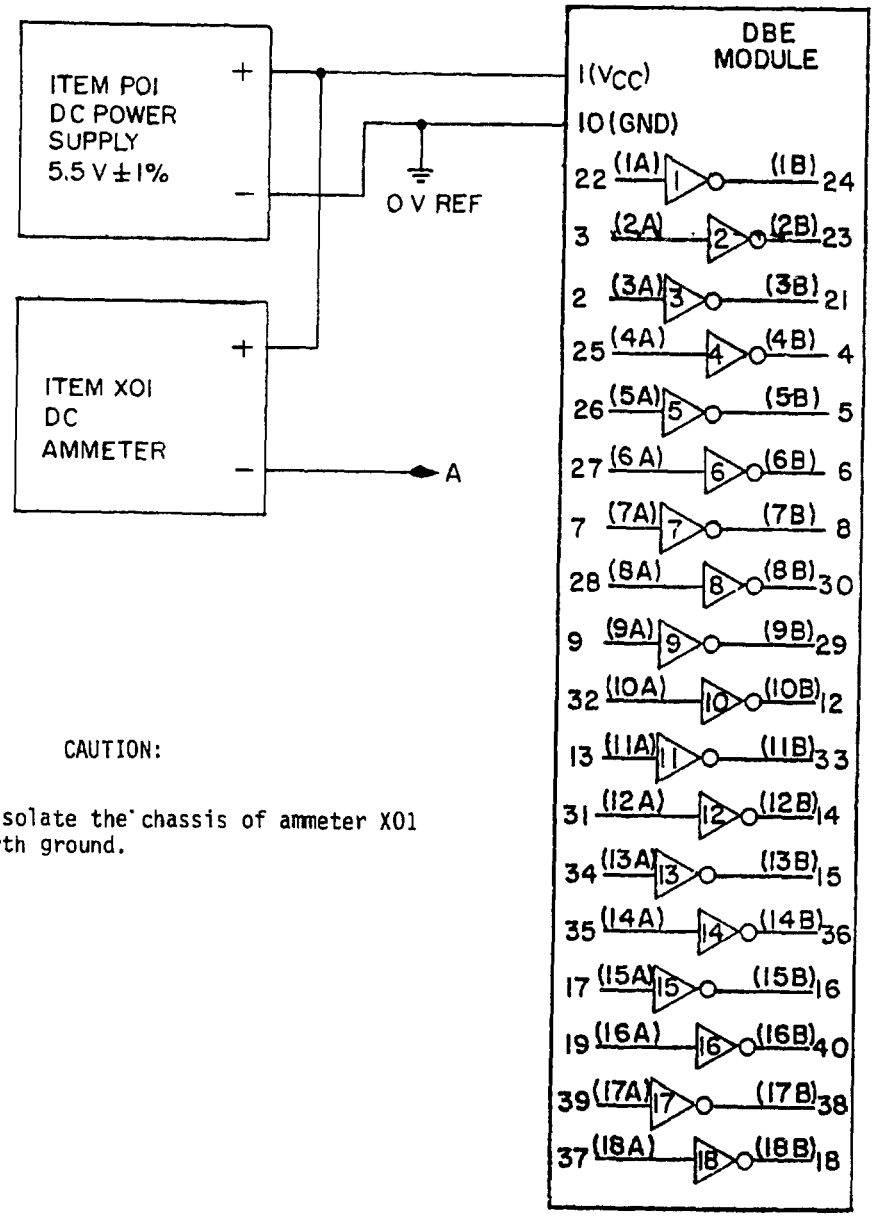
FIGURE 5 V_{OUT} (low-loaded) test circuit for table III.

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TABLE IV. Input breakdown current test.

Test condition	Parameter	Connect	Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
B1	1A Input breakdown current	22	0	100	µA	6
B2	2A	3	↓	↓	↓	↓
B3	3A	2				
B4	4A	25				
B5	5A	26				
B6	6A	27				
B7	7A	7				
B8	8A	28				
B9	9A	9				
B10	10A	32				
B11	11A	13				
B12	12A	31				
B13	13A	34				
B14	14A	35				
B15	15A	17				
B16	16A	19				
B17	17A	39				
B18	18A	37				

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

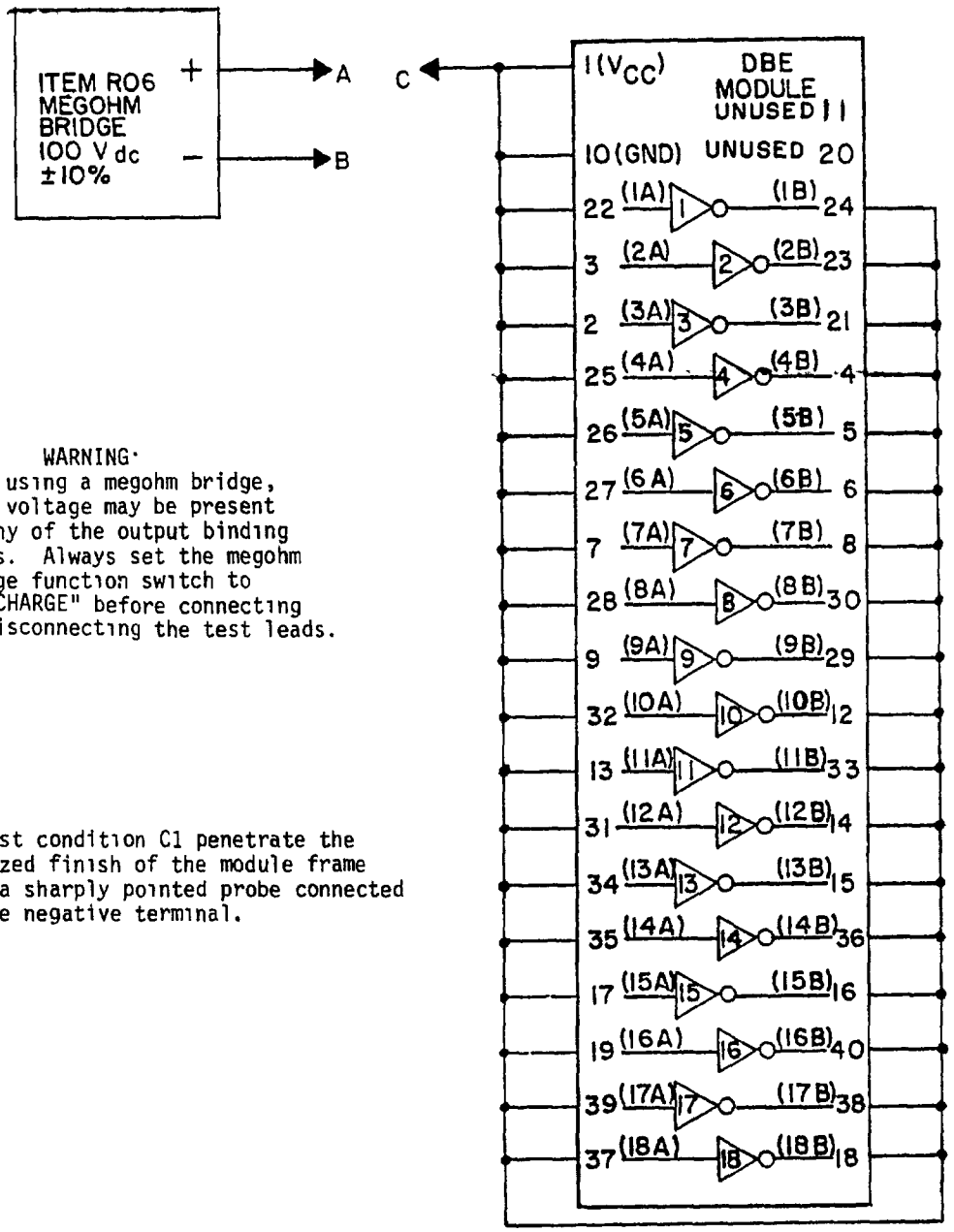
FIGURE 6. Input breakdown current test circuit for table IV.

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TABLE V. Isolation test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A pin	B pin	Min	Max		
C1	Isolation, frame to module pins	C,11,20	Frame	10	∞	MΩ	7
C2	Isolation, used to unused pins	A	11,20	10	∞	MΩ	7
C3	Isolation, unused to unused pins	11	20	10	∞	MΩ	7

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WARNING:
When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "DISCHARGE" before connecting or disconnecting the test leads.

NOTE:
On test condition C1 penetrate the anodized finish of the module frame with a sharply pointed probe connected to the negative terminal.

FIGURE 7. Isolation test circuit for table V.

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TABLE VI. Output voltage (low-level) test.

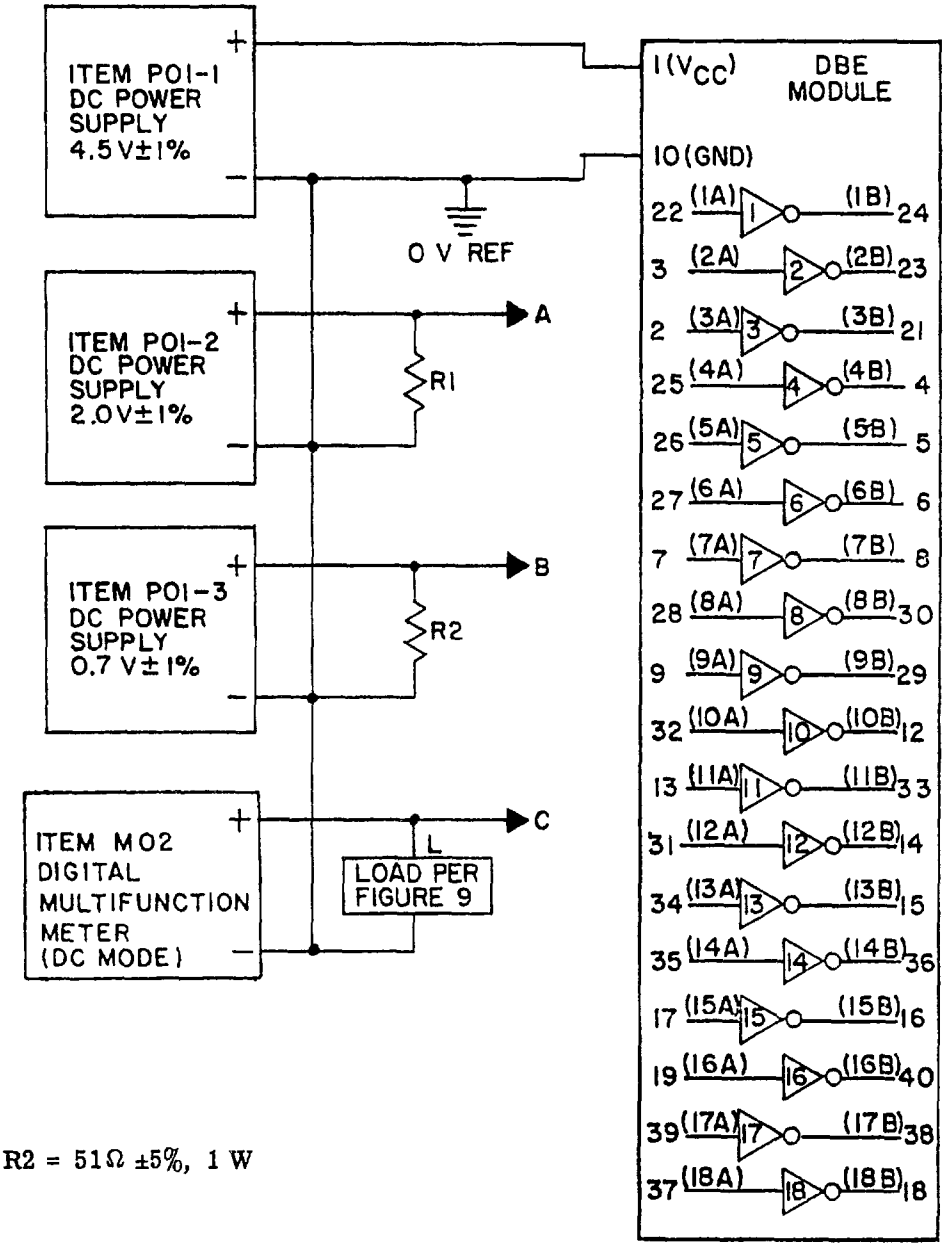
Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
D1	1B Output voltage, low level	22	Open	24	0.0	0.3	V	8
D2	2B	3	↓	23	↓	↓	↓	↓
D3	3B	2		21				
D4	4B	25		4				
D5	5B	26		5				
D6	6B	27		6				
D7	7B	7		8				
D8	8B	28		30				
D9	9B	9		29				
D10	10B	32		12				
D11	11B	13		33				
D12	12B	31		14				
D13	13B	34		15				
D14	14B	35		36				
D15	15B	17		16				
D16	16B	19		40				
D17	17B	39		38				
D18	18B	37		18				

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TABLE VII. Output voltage (high-level) test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
E1	1B Output voltage, high level	Open	22	24	2.4	4.5	V	8
E2	2B		3	23				
E3	3B		2	21				
E4	4B		25	4				
E5	5B		26	5				
E6	6B		27	6				
E7	7B		7	8				
E8	8B		28	30				
E9	9B		9	29				
E10	10B		32	12				
E11	11B		13	33				
E12	12B		31	14				
E13	13B		34	15				
E14	14B		35	36				
E15	15B		17	16				
E16	16B		19	40				
E17	17B		39	38				
E18	18B		37	18				

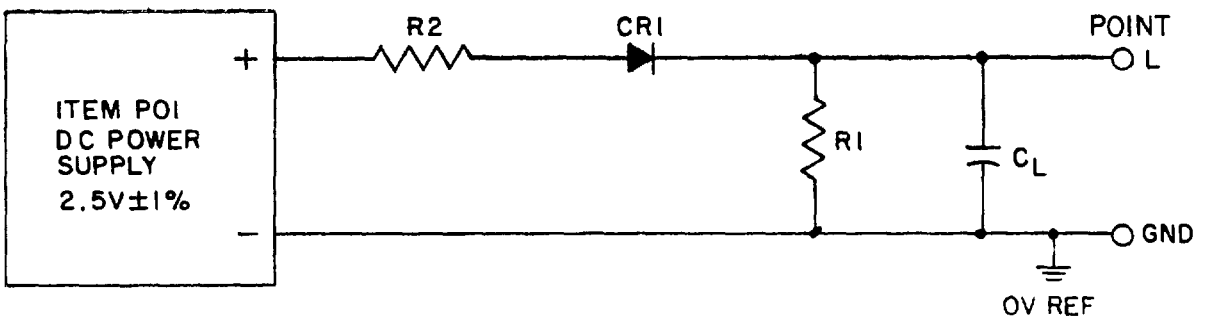
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NOTE: R1 = R2 = 51Ω ±5%, 1 W

FIGURE 8. Output voltage (low-level and high-level) test circuit for table VI and VII.

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NOTES
1. CR1 = 1N916 or equivalent.
2. R1 = 23.7 kΩ ±1%, 1/8 W (current flow of 100 µA when point L is held at 2.4 V).
3. R2 = Select value for a current flow of 2.0 mA ±5% (when point L is held at 0.3 V).
4. C_L = 40 ±5 pF (including probe capacitance) for test conditions K1-K18, L1-L18, M1-M18 and N1-N18.
   = 140 ±5 pF (including probe capacitance) for test conditions K19-K36, L19-L36, M19-M36 and N19-N36
   = 50 ±5 pF (including probe capacitance) for test conditions P1-P18 and R1-R18.
5. C_L is not required for dc electrical test conditions D and E (table II).
6. The maximum lead length from point "L" to the appropriate module pin shall be less than 1 inch (25.4 mm)
7. During output switching, the 2.5 V power supply shall have a maximum overshoot of 500 mV.

FIGURE 9. Logic load circuit.

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TABLE VIII. Input current (low-level) test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
F1	1A Input current, low level	Open	22	Open	50	200	µA	10
F2	2A	↓	3	↓	↓	↓	↓	↓
F3	3A		2					
F4	4A		25					
F5	5A		26					
F6	6A		27					
F7	7A		7					
F8	8A		28					
F9	9A		9					
F10	10A		32					
F11	11A		13					
F12	12A		31					
F13	13A		34					
F14	14A		35					
F15	15A		17					
F16	16A		19					
F17	17A		39					
F18	18A		37					

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TABLE IX. Input current (high-level) test.

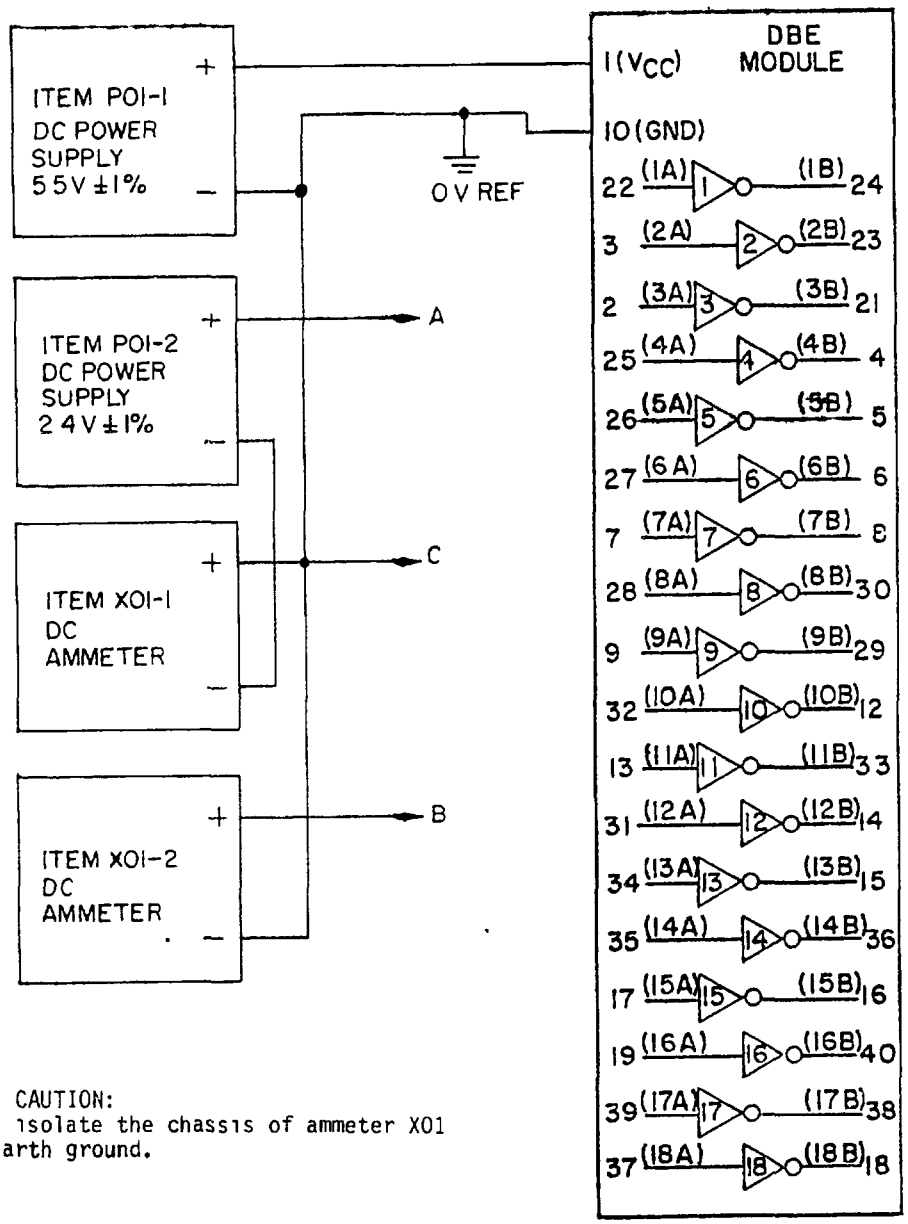
Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
G1	1A Input current, high level	22	Open	Open	0	10	µA	10
G2	2A	3						
G3	3A	2						
G4	4A	25						
G5	5A	26						
G6	6A	27						
G7	7A	7						
G8	8A	28						
G9	9A	9						
G10	10A	32						
G11	11A	13						
G12	12A	31						
G13	13A	34						
G14	14A	35						
G15	15A	17						
G16	16A	19						
G17	17A	39						
G18	18A	37						

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TABLE X. Output short circuit current test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
H1	1B Output short circuit current	Open	24	22	3	15	mA	10
H2	2B		23	3				
H3	3B		21	2				
H4	4B		4	25				
H5	5B		5	26				
H6	6B		6	27				
H7	7B		8	7				
H8	8B		30	28				
H9	9B		29	9				
H10	10B		12	32				
H11	11B		33	13				
H12	12B		14	31				
H13	13B		15	34				
H14	14B		36	35				
H15	15B		16	17				
H16	16B		40	19				
H17	17B		38	39				
H18	18B		18	37				

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CAUTION:
Do not isolate the chassis of ammeter XO1 from earth ground.

NOTES:
1. Since items XO1-1 and XO1-2 are not used simultaneously, only one item XO1 is required.
2. With point "A" disconnected, item XO1-1 should indicate zero current on the 1 µA range.
3. When performing output short circuit test, only one output shall be shorted at a time and the measurement taken within 5 seconds.

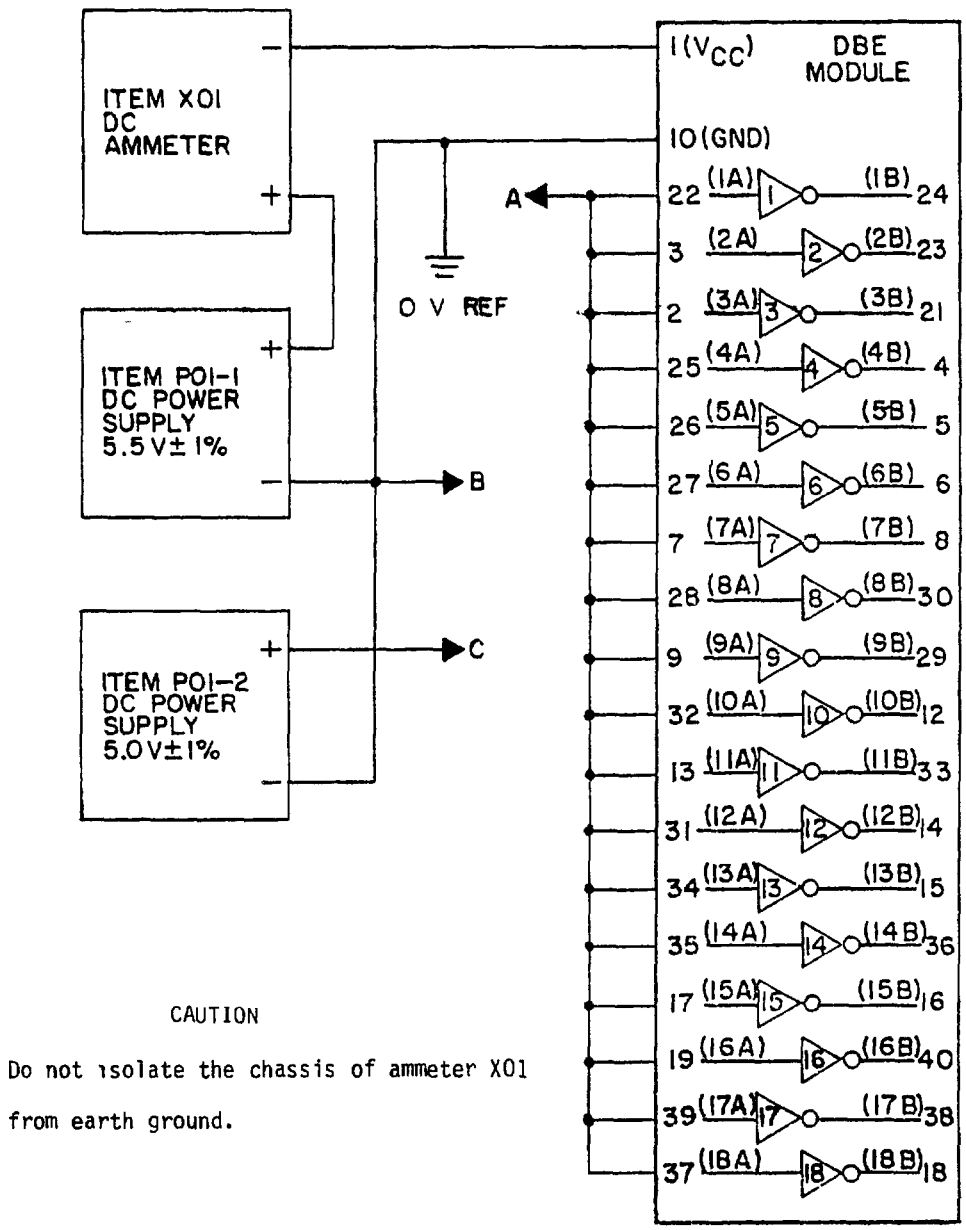
FIGURE 10. Input current (low-level and high-level) and output short circuit current test circuit for tables VIII, IX, and X.

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TABLE XI. Power supply current (low-level and high-level) test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
J1	Power supply current, low level	C	Open	A	1.0	9.2	mA	11
J2	Power supply current, high level	B	A	Open	0.4	3.6	mA	11

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CAUTION

Do not isolate the chassis of ammeter XO1 from earth ground.

FIGURE 11. Power supply current test circuit for table XI.

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TABLE XII. Transition time (t_{TLH}) test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
K1	1B Transition time (t_{TLH}), C_L = 40 pF	22	24	5	60	ns	12,13
K2	2B	3	23	↓	↓	↓	↓
K3	3B	2	21				
K4	4B	25	4				
K5	5B	26	5				
K6	6B	27	6				
K7	7B	7	8				
K8	8B	28	30				
K9	9B	9	29				
K10	10B	32	12				
K11	11B	13	33				
K12	12B	31	14				
K13	13B	34	15				
K14	14B	35	36				
K15	15B	17	16				
K16	16B	19	40				
K17	17B	39	38				
K18	18B	37	18				
K19	1B Transition time (t_{TLH}), C_L = 140 pF	22	24	10	100		
K20	2B	3	23	↓	↓		
K21	3B	2	21				
K22	4B	25	4				
K23	5B	26	5				
K24	6B	27	6				
K25	7B	7	8				
K26	8B	28	30				
K27	9B	9	29				
K28	10B	32	12				
K29	11B	13	33				
K30	12B	31	14				
K31	13B	34	15				
K32	14B	35	36				
K33	15B	17	16				
K34	16B	19	40				
K35	17B	39	38				
K36	18B	37	18				

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TABLE XIII. Transition time (t_{THL}) test.

Test condition	Parameter	Connect A to pin	Connect B to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
L1	1B Transition time (t_{THL}), C_L = 40 pF	22	24	2	30	ns	12,13
L2	2B	3	23				
L3	3B	2	21				
L4	4B	25	4				
L5	5B	26	5				
L6	6B	27	6				
L7	7B	7	8				
L8	8B	28	30				
L9	9B	9	29				
L10	10B	32	12				
L11	11B	13	33				
L12	12B	31	14				
L13	13B	34	15				
L14	14B	35	36				
L15	15B	17	16				
L16	16B	19	40				
L17	17B	39	38				
L18	18B	37	18				
L19	1B Transition time (t_{THL}), C_L = 140 pF	22	24	5	60		
L20	2B	3	23				
L21	3B	2	21				
L22	4B	25	4				
L23	5B	26	5				
L24	6B	27	6				
L25	7B	7	8				
L26	8B	28	30				
L27	9B	9	29				
L28	10B	32	12				
L29	11B	13	33				
L30	12B	31	14				
L31	13B	34	15				
L32	14B	35	36				
L33	15B	17	16				
L34	16B	19	40				
L35	17B	39	38				
L36	18B	37	18				

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TABLE XIV. Propagation delay time (t_{PLH}) test.

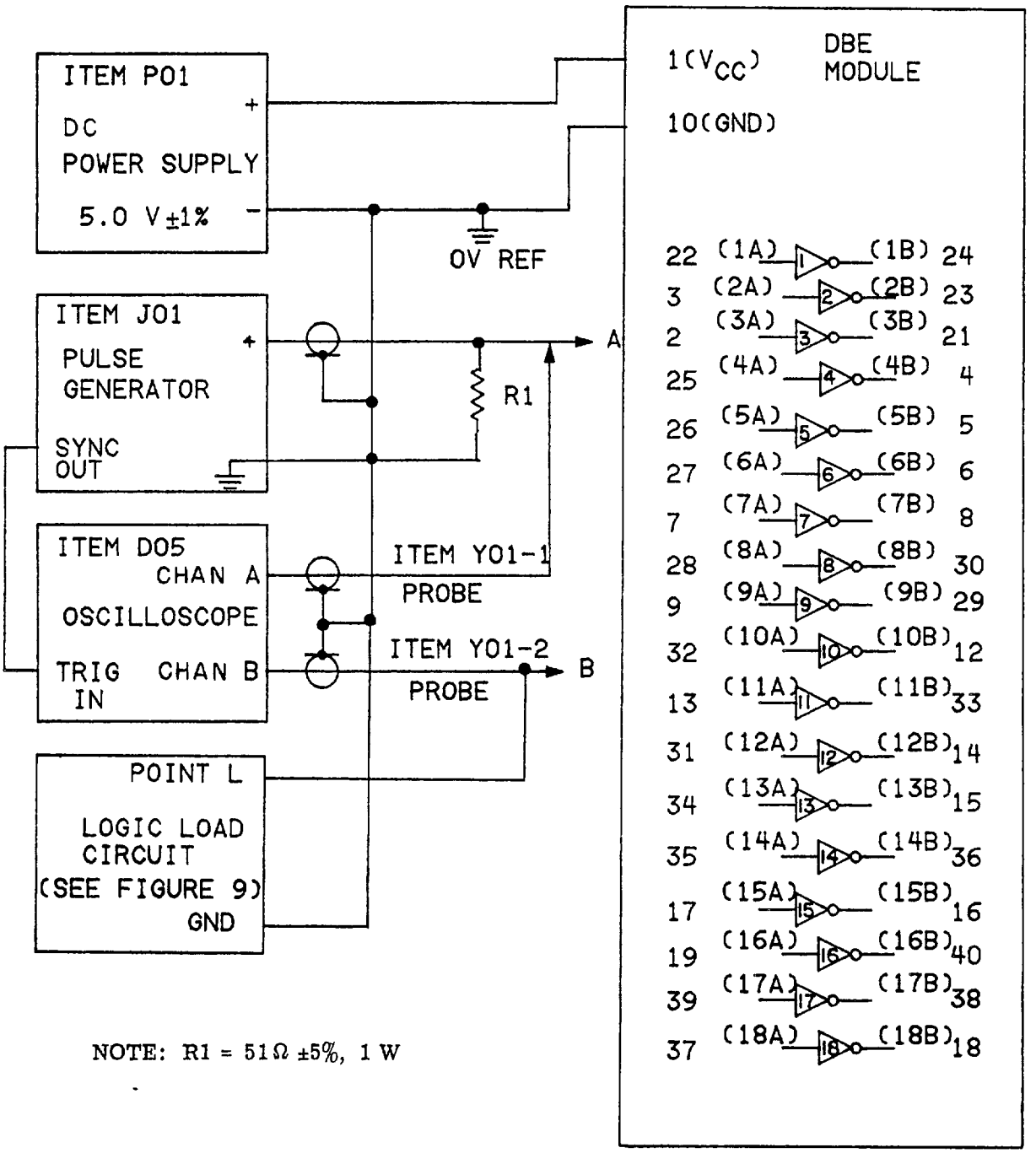
Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
M1	1B Propagation delay time (t_{PLH}), C_L = 40 pF	22	24	10	60	ns	12,13
M2	2B	3	23	↓	↓	↓	↓
M3	3B	2	21				
M4	4B	25	4				
M5	5B	26	5				
M6	6B	27	6				
M7	7B	7	8				
M8	8B	28	30				
M9	9B	9	29				
M10	10B	32	12				
M11	11B	13	33				
M12	12B	31	14				
M13	13B	34	15				
M14	14B	35	36				
M15	15B	17	16				
M16	16B	19	40				
M17	17B	39	38				
M18	18B	37	18				
M19	1B Propagation delay time (t_{PLH}), C_L = 140 pF	22	24	10	90		
M20	2B	3	23	↓	↓		
M21	3B	2	21				
M22	4B	25	4				
M23	5B	26	5				
M24	6B	27	6				
M25	7B	7	8				
M26	8B	28	30				
M27	9B	9	29				
M28	10B	32	12				
M29	11B	13	33				
M30	12B	31	14				
M31	13B	34	15				
M32	14B	35	36				
M33	15B	17	16				
M34	16B	19	40				
M35	17B	39	38				
M36	18B	37	18				

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TABLE XV. Propagation delay time (t_{PHL}) test.

Test condition	Parameter	Connect A to pin	Connect B to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
N1	1B Propagation delay time (t_{PHL}), C_L = 40 pF	22	24	10	66	ns	12,13
N2	2B	3	23				
N3	3B	2	21				
N4	4B	25	4				
N5	5B	26	5				
N6	6B	27	6				
N7	7B	7	8				
N8	8B	28	30				
N9	9B	9	29				
N10	10B	32	12				
N11	11B	13	33				
N12	12B	31	14				
N13	13B	34	15				
N14	14B	35	36				
N15	15B	17	16				
N16	16B	19	40				
N17	17B	39	38				
N18	18B	37	18				
N19	1B Propagation delay time (t_{PHL}), C_L = 140 pF	22	24	10	96		
N20	2B	3	23				
N21	3B	2	21				
N22	4B	25	4				
N23	5B	26	5				
N24	6B	27	6				
N25	7B	7	8				
N26	8B	28	30				
N27	9B	9	29				
N28	10B	32	12				
N29	11B	13	33				
N30	12B	31	14				
N31	13B	34	15				
N32	14B	35	36				
N33	15B	17	16				
N34	16B	19	40				
N35	17B	39	38				
N36	18B	37	18				

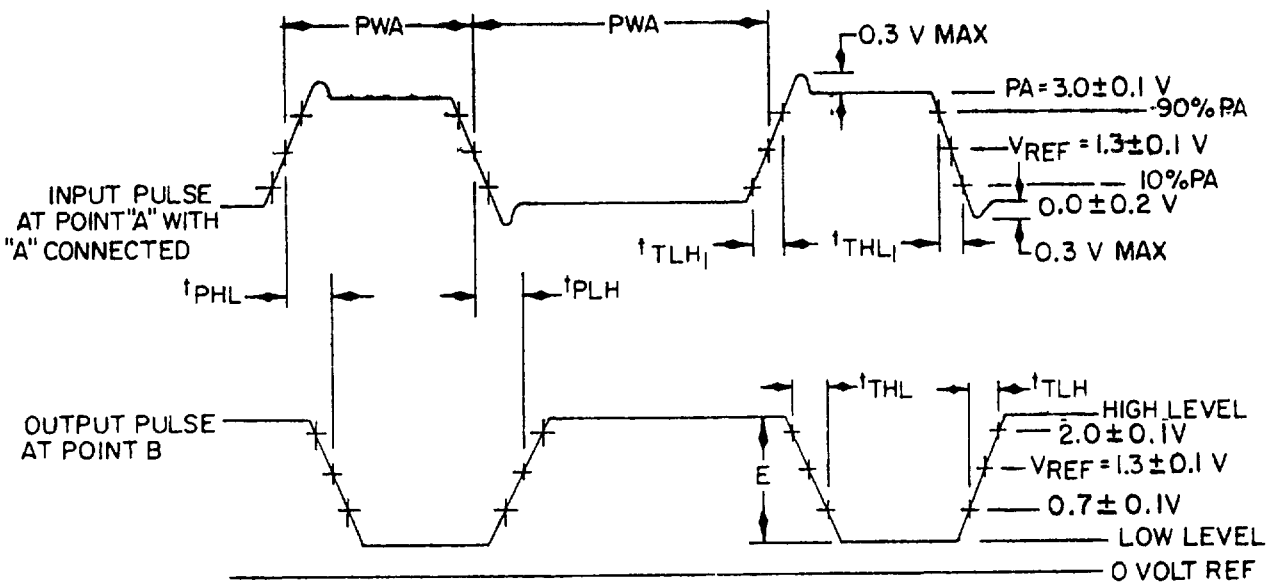
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NOTE: R1 = 51Ω ±5%, 1 W

FIGURE 12. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit for tables XII through XV.

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NOTES

1. Input pulse parameters:

   $t_{TLH_1} = t_{THL_1} = 10 \pm 2$ ns.
   PWA = 500 ±50 ns.

2. Before making the measurements verify that the output pulse high level is $\geq$ 2.4 V and the low level is $\leq$ 0.3 V.

3. All voltages are referenced to ground.

FIGURE 13. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for tables XII through XV.

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TABLE XVI. Output crosstalk (low-level) test.

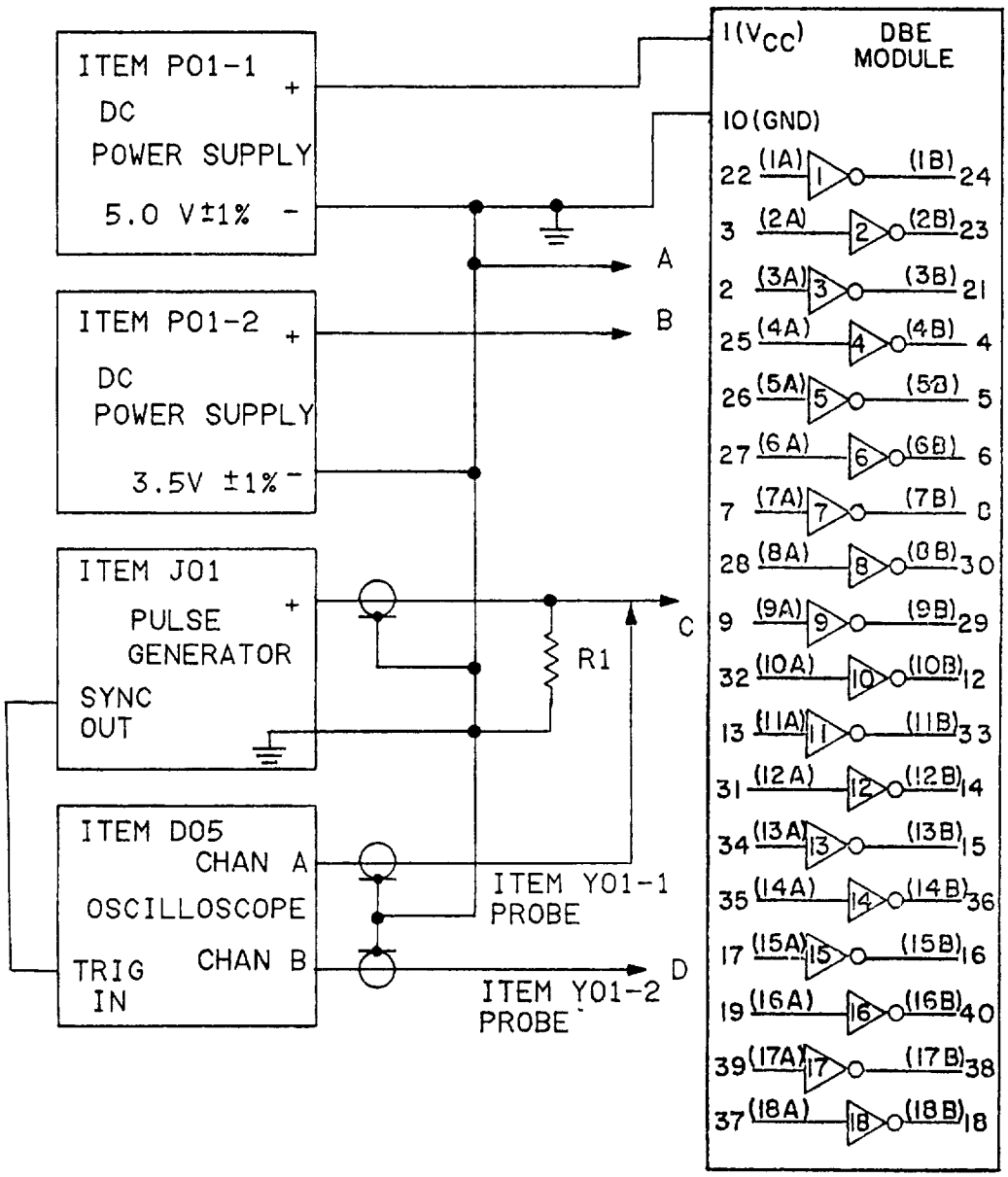
Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P1	1B Output crosstalk, low level	Open	22	3,2,25,26,27, 7,28, 9,32,13,31,34,35,17,19.39,37	24	0	400	mV	14,15
P2	2B		3	22,2,25,26,27, 7,28, 9,32,13,31,34,35,17,19,39,37	23				
P3	3B		2	22,3,25,26,27, 7,28, 9,32,13,31,34,35,17,19,39,37	21				
P4	4B		25	22,3, 2,26,27, 7,28, 9,32,13,31,34,35,17,19,39,37	4				
P5	5B		26	22,3, 2,25,27, 7,28, 9,32,13,31,34,35,17,19,39,37	5				
P6	6B		27	22,3, 2,25,26, 7,28, 9,32,13,31,34,35,17,19,39,37	6				
P7	7B		7	22,3, 2,25,26,27,28, 9,32,13,31,34,35,17,19,39,37	8				
P8	8B		28	22,3, 2,25,26,27, 7, 9,32,13,31,34,35,17,19,39,37	30				
P9	9B		9	22,3, 2,25,26,27, 7,28,32,13,31,34,35,17,19,39,37	29				
P10	10B		32	22,3, 2,25,26,27, 7,28, 9,13,31,34,35,17,19,39,37	12				
P11	11B		13	22,3, 2,25,26,27, 7,28, 9,32,31,34,35,17,19,39,37	33				
P12	12B		31	22,3, 2,25,26,27, 7,28, 9,32,13,34,35,17,19,39,37	14				
P13	13B		34	22,3, 2,25,26,27, 7,28, 9,32,13,31,35,17,19,39,37	15				
P14	14B		35	22,3, 2,25,26,27, 7,28, 9,32,13,31,34,17,19,39,37	36				
P15	15B		17	22,3, 2,25,26,27, 7,28, 9,32,13,31,34,35,19,39,37	16				
P16	16B		19	22,3, 2,25,26,27, 7,28, 9,32,13,31,34,35,17,39,37	40				
P17	17B		39	22,3, 2,25,26,27, 7,28, 9,32,13,31,34,35,17,19,37	38				
P18	18B		37	22,3, 2,25,26,27, 7,28, 9,32,13,31,34,35,17,19,39	18				

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TABLE XVII. Output crosstalk (high-level) test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
R1	1B Output crosstalk, high level	22	Open	3,2,25,26,27, 7,28, 9,32,13,31,34,35,17,19,39,37	24	0	400	mV	14, 15
R2	2B	3		22,2,25,26,27, 7,28, 9,32,13,31,34,35,17,19,39,37	23				
R3	3B	2		22,3,25,26,27, 7,28, 9,32,13,31,34,35,17,19,39,37	21				
R4	4B	25		22,3, 2,26,27, 7,28, 9,32,13,31,34,35,17,19,39,37	4				
R5	5B	26		22,3, 2,25,27, 7,28, 9,32,13,31,34,35,17,19,39,37	5				
R6	6B	27		22,3, 2,25,26, 7,28, 9,32,13,31,34,35,17,19,39,37	6				
R7	7B	7		22,3, 2,25,26,27,28, 9,32,13,31,34,35,17,19,39,37	8				
R8	8B	28		22,3, 2,25,26,27, 7, 9,32,13,31,34,35,17,19,39,37	30				
R9	9B	9		22,3, 2,25,26,27, 7,28,32,13,31,34,35,17,19,39,37	29				
R10	10B	32		22,3, 2,25,26,27, 7,28, 9,13,31,34,35,17,19,39,37	12				
R11	11B	13		22,3, 2,25,26,27, 7,28, 9,32,31,34,35,17,19,39,37	33				
R12	12B	31		22,3, 2,25,26,27, 7,28, 9,32,13,34,35,17,19,39,37	14				
R13	13B	34		22,3, 2,25,26,27, 7,28, 9,32,13,31,35,17,19,39,37	15				
R14	14B	35		22,3, 2,25,26,27, 7,28, 9,32,13,31,34,17,19,39,37	36				
R15	15B	17		22,3, 2,25,26,27, 7,28, 9,32,13,31,34,35,19,39,37	16				
R16	16B	19		22,3, 2,25,26,27, 7,28, 9,32,13,31,34,35,17,39,37	40				
R17	17B	39		22,3, 2,25,26,27, 7,28, 9,32,13,31,34,35,17,19,37	38				
R18	18B	37		22,3, 2,25,26,27, 7,28, 9,32,13,31,34,35,17,19,39	18				

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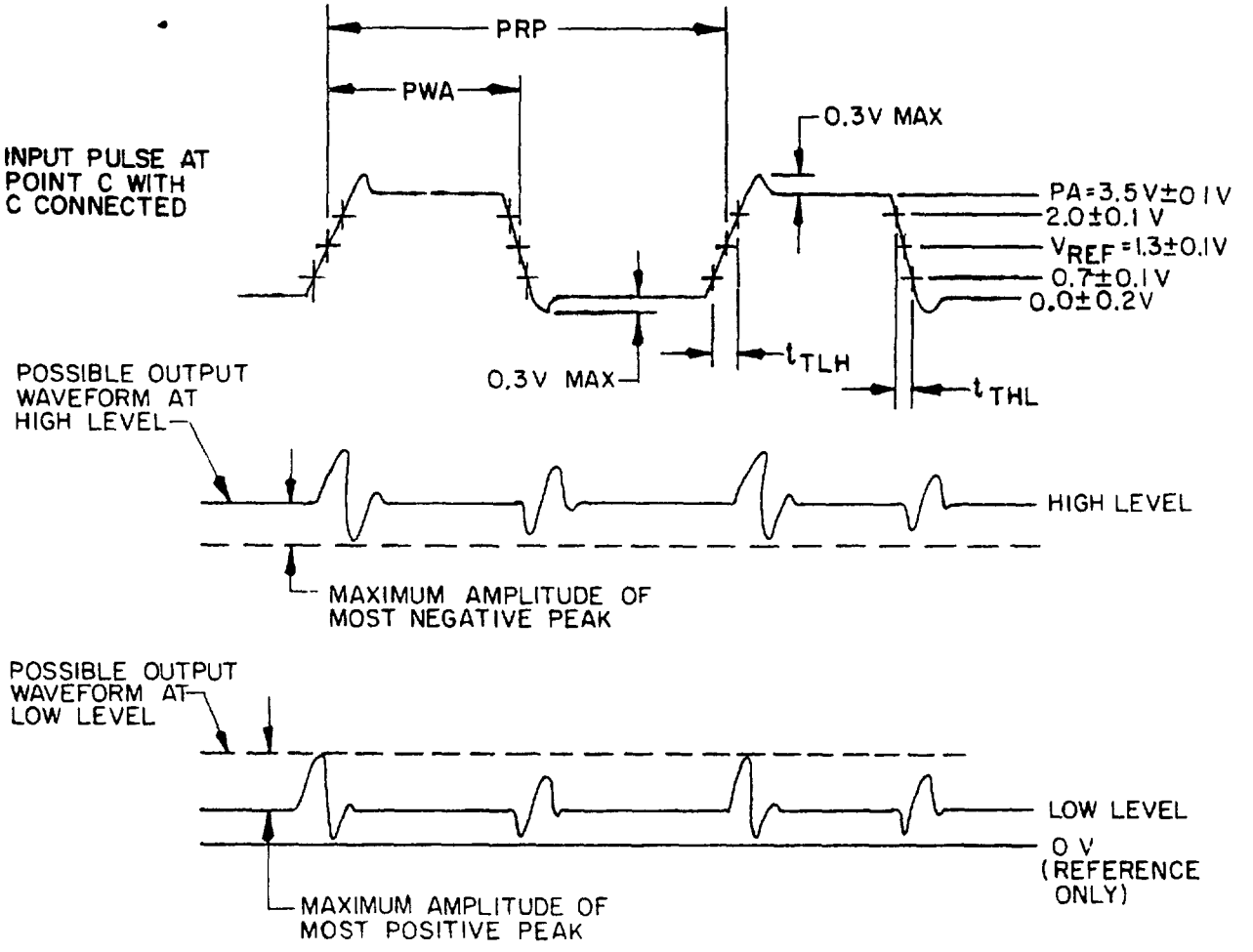


NOTES:

1. R1 = 51Ω ±5%, 1W.
2. Load all outputs with the load specified on figure 9.

FIGURE 14. Output crosstalk (low-level and high-level) test circuit for tables XVI and XVII

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NOTE:
Input pulse parameters.
$t_{THL} = t_{TLH} = 3.5 \pm 1.5$ ns (verified at all input pins)
PWA = 200 ±20 ns.
PRP = 1000 ±100 ns.

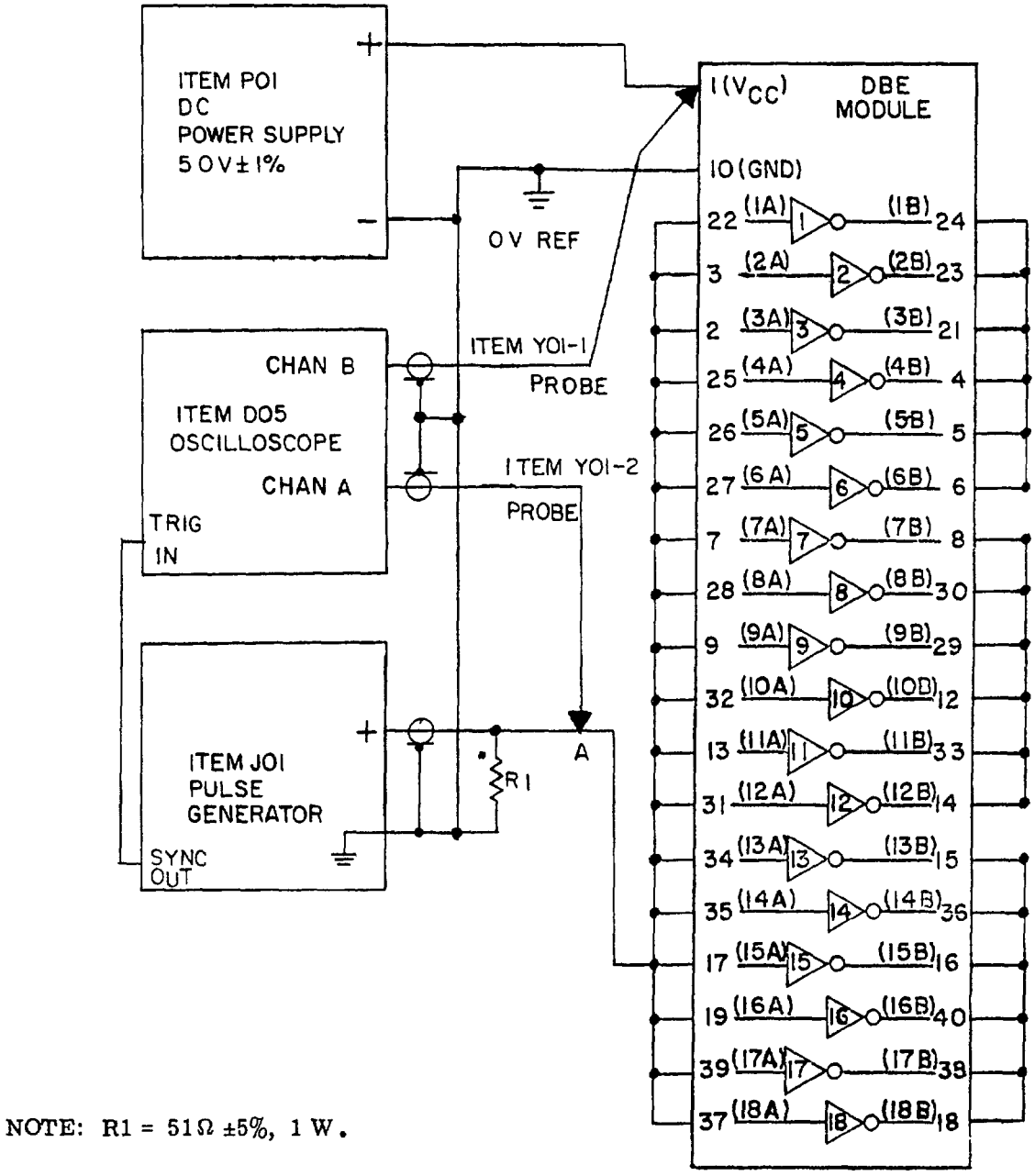
.FIGURE 15. Output crosstalk (low-level and high-level) test circuit waveforms for tables XVI and XVII

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TABLE XVIII. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
S	Power supply transient amplitude	0	300	mV	16,17

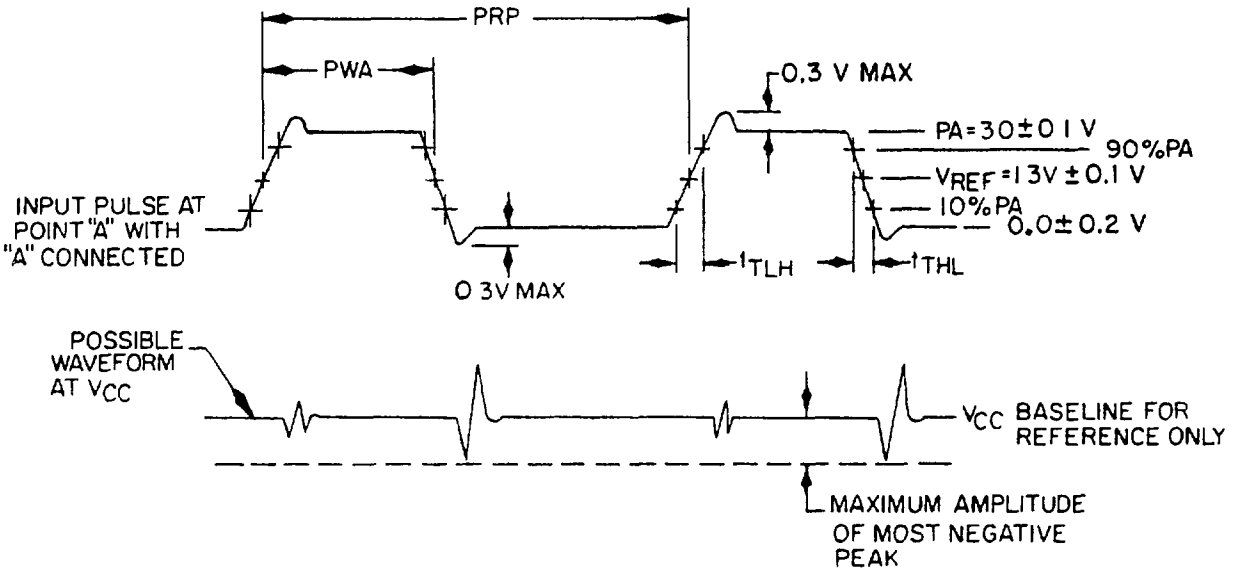
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NOTE: R1 = 51Ω ±5%, 1 W.

FIGURE 16. Power supply transient amplitude test circuit for table XVIII.

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NOTE:
Input pulse parameters:
$t_{TLH} = t_{THL} = 20 \pm 2$ ns.
PWA = 200 ±20 ns.
PRP = 1000 ±100 ns.

FIGURE 17. Power supply transient amplitude test circuit waveforms for table XVIII

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TABLE XIX. Power supply filter capacitance test.

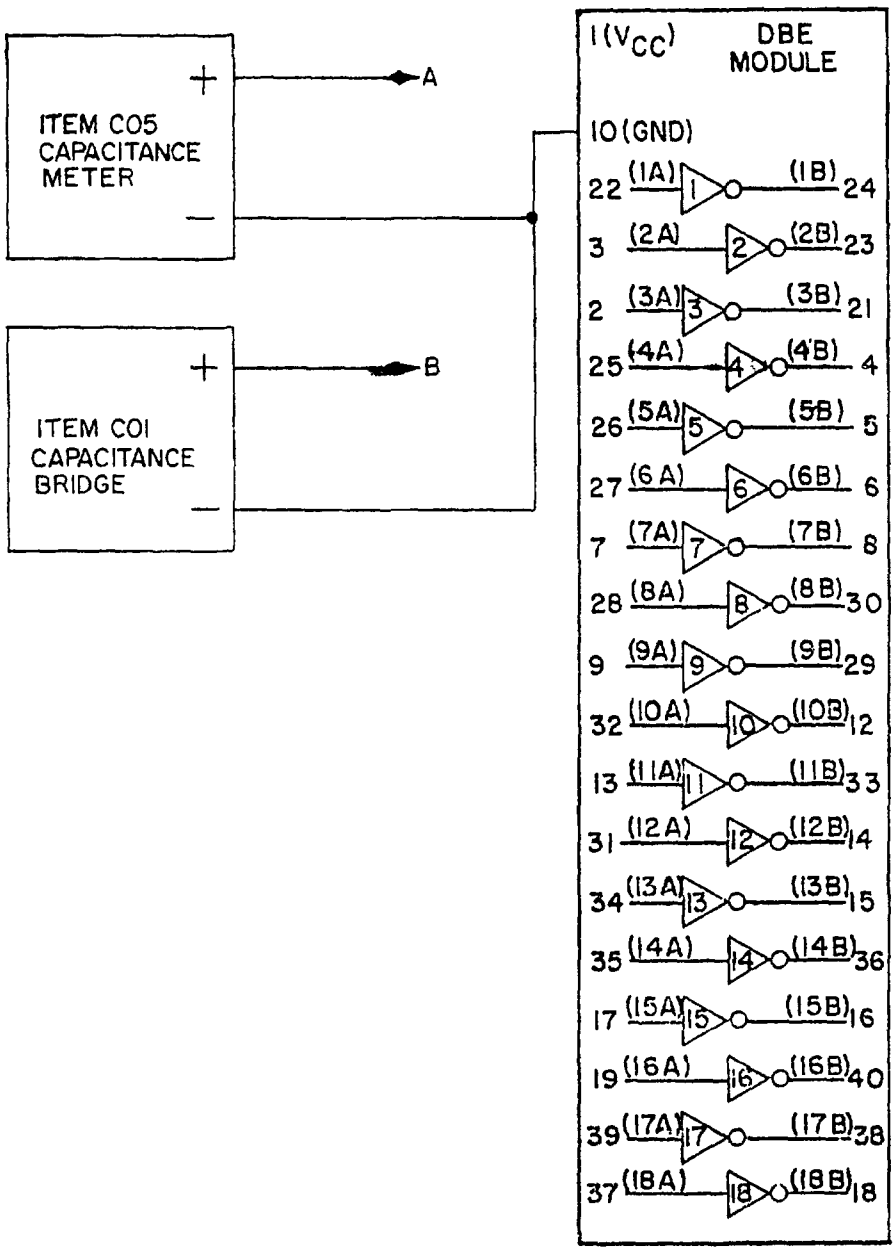
Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
T	Power supply filter capacitance	Open	1	1.65	2.75	µF	18

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TABLE XX. Input capacitance test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
U1	1A Input capacitance	22	Open	2	15	pF	18
U2	2A	3	↓	↓	↓	↓	↓
U3	3A	2					
U4	4A	25					
U5	5A	26					
U6	6A	27					
U7	7A	7					
U8	8A	28					
U9	9A	9					
U10	10A	32					
U11	11A	13					
U12	12A	31					
U13	13A	34					
U14	14A	35					
U15	15A	17					
U16	16A	19					
U17	17A	39					
U18	18A	37					

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NOTES:
1. Null out the effect of the test leads prior to connecting to the module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.
3. Capacitance bridge (C01) set-up:
   Frequency = 120 Hz.
   V_{pk} = 0.3 V max.

FIGURE 18. Power supply filter and input capacitance test circuit for tables XIX and XX.

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TABLE XXI. Output impedance (power off) check test.

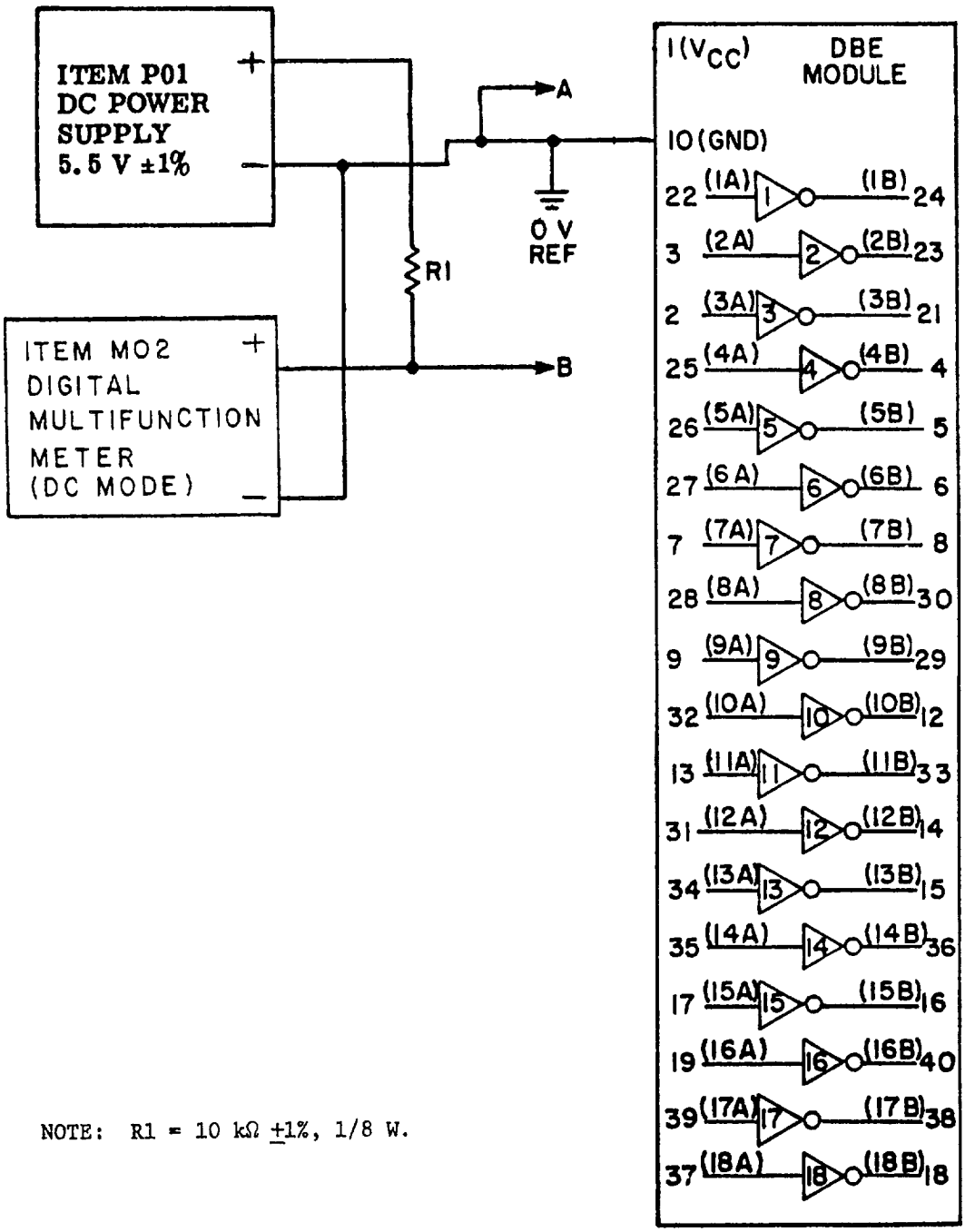
Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
V1	1B Output impedance, power off, check	Open	24	3.5	5.55	V	19
V2	2B		23				
V3	3B		21				
V4	4B		4				
V5	5B		5				
V6	6B		6				
V7	7B		8				
V8	8B		30				
V9	9B		29				
V10	10B		12				
V11	11B		33				
V12	12B		14				
V13	13B		15				
V14	14B		36				
V15	15B		16				
V16	16B		40				
V17	17B		38				
V18	18B		18				

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TABLE XXI. Output impedance (power off) check test. - Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
V19	1B Output impedance, power off, check	1	24	3.5	5.55	V	19
V20	2B		23				
V21	3B		21				
V22	4B		4				
V23	5B		5				
V24	6B		6				
V25	7B		8				
V26	8B		30				
V27	9B		29				
V28	10B		12				
V29	11B		33				
V30	12B		14				
V31	13B		15				
V32	14B		36				
V33	15B		16				
V34	16B		40				
V35	17B		38				
V36	18B		18				

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NOTE: R1 = 10 kΩ ±1%, 1/8 W.

FIGURE 19. Output impedance (power off) check test circuit for table XXI.

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